

Reviewer Pack — Minimal Order of Hodge Decomposition and SAT Clause Subset E...

Entry 84da68a6c9a7 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Order of Hodge Decomposition and SAT Clause Subset Entropy Correlation

Entry ID: 84da68a6c9a7 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-04 21:39:20 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Hodge Decomposition) **Field B** (complexity object): Boolean Satisfiability (SAT Clause Subsets)

Statement:

For any CNF formula φ with n variables, the minimal order of the Hodge decomposition of its corresponding complex projective variety is linearly correlated with its clause subset entropy, such that $H(\varphi) = \Theta(O(n^{1.5}))$, where $H(\varphi)$ denotes the Hodge decomposition order and $O(n^{1.5})$ is a function of the number of clauses.

Rationale (proposer's reasoning):

Hodge Decomposition provides a way to analyze the geometric structure of complex varieties, which may reveal hidden complexities in the clause subsets of SAT formulas. A correlation between these two measures could provide insights into the structure of SAT solvers and potentially lead to new algorithms with improved complexity bounds.

Taxonomy category: AlgebraicGeometry × BooleanSatisfiability (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): ce7d47355dd594a3

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For the given CNF formula φ with n variables, we support the conjecture if the correlation coefficient between the Hodge decomposition order and clause subset entropy is $|r| \geq 0.8$ for at least 24 out of 30 seeds, and no seed produces a correlation coefficient < 0.5 .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Hodge Decomposition" AND "SAT Clause Subsets" AND Correlation - "Algebraic Geometry" AND "Boolean Satisfiability" AND Hodge decomposition order - entropy AND CNF formula AND complex projective variety

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def generate_cnf(n, m):
    cnf = []
    for _ in range(m):
        clause = [random.randint(1, n), -random.randint(1, n)]
        if random.choice([True, False]):
            clause[0] *= -1
        if random.choice([True, False]):
            clause[1] *= -1
        cnf.append(clause)
    return cnf

def clause_subset_entropy(cnf):
    total_clauses = len(cnf)
    subset_sizes = [len(subset) for subset in range(1, total_clauses + 1)]
    counts = [0] * (total_clauses + 1)

    for i in range(1 << total_clauses):
        subset = []
        for j in range(total_clauses):
            if i & (1 << j):
                subset.append(cnf[j])
        counts[len(subset)] += 1

    probabilities = [count / (2 ** total_clauses) for count in counts]
    entropy = -sum(p * math.log2(p) for p in probabilities if p > 0)
    return entropy

def hodge_decomposition_order(n):
    # Placeholder function to simulate Hodge decomposition order
    # This is a dummy implementation and should be replaced with actual logic
```

```

    return n ** 1.5

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = random.randint(5, 40)
    m = random.randint(n, n * 3)
    cnf = generate_cnf(n, m)

    hodge_order = hodge_decomposition_order(n)
    entropy = clause_subset_entropy(cnf)

    correlation_coefficient = (hodge_order - n ** 1.5) / (n ** 2.5 - n ** 1.5)

    return {
        "metric_name": "correlation_coefficient",
        "metric_value": correlation_coefficient,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": abs(correlation_coefficient) >= 0.8,
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(arg) for arg in sys.argv[1:]] or [random.randint(2, 1000000) for _ in range(30)]

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(result["metric_value"] for result in results) / len(results)
    std_value = math.sqrt(sum((result["metric_value"] - mean_value) ** 2 for result in results) / len(results))
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not result["conjecture_holds"] for result in results) and sum(1 for result in results if result["conjecture_holds"]) > 0:
        print(f"RESULT: FALSIFIED counterexample=\"not_enough_failing_seeds\" first_failing_seed={seeds[0]}")
    else:
        print(f"RESULT: INCONCLUSIVE support_fraction={support_fraction}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_24f2c758.py", line 77, in <module>
    trial_result = run_trial(seed)
                   ~~~~~^~~~~~

```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_24f2c758.py", line 58, in run_trial
    entropy = clause_subset_entropy(cnf)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_24f2c758.py", line 31, in clause_subset
    subset_sizes = [len(subset) for subset in range(1, total_clauses + 1)]
    ^^^^^^^^^^^
```

TypeError: object of type 'int' has no len()

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which means it did not complete its execution to provide a correlation coefficient for the conjecture. | next: Review and debug the test code to ensure it can run to completion and produce the required data.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	15232
2	preregistration	ollama_remote	glm4:latest	0	0	9395
3	novelty	ollama_remote	glm4:latest	0	0	17713
4	novelty	ollama_remote	glm4:latest	0	0	8893
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13806
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9869
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13542
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9249
9	judge	ollama_remote	glm4:latest	0	0	11962

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 109661 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/84da68a6c9a7.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/84da68a6c9a7.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/84da68a6c9a7.tar.gz (if generated)